

PRICE SIM · Mathematics Project

Real Estate **Price Simulator**

Correlation & Multiple Regression applied to 2,518 real Hyderabad property listings — a web application with an AI assistant that prices homes in Indian rupees.

Group: Class 206 · Department of Mathematics · 2025–26

Guide: LG No. 11 · **Members:** Bhairavesh (4160) · Irfan (4161) · Rahul (4169) · Krishna (4165) · Pruthvi Raj (4157)

1. Introduction & Objective

Buying a home is the single largest financial decision most people make. In Hyderabad, the price of a flat can differ by more than **2.5x** with the same area and number of rooms, purely because of the locality — a 1,500 sq ft 3-BHK costs about ₹64 L in Nizampet but ₹1.63 Cr in Banjara Hills. This project asks a simple mathematical question: **how do area, rooms, amenities and location drive the price of a home, and can we quantify that relationship?**

The project answers it with the statistical toolkit of a first course in regression: **Pearson correlation** to measure individual relationships, and **multiple linear regression** to model all factors together. The fitted models are deployed as an interactive web application so the mathematics can be experienced live: drag a slider, pick a locality, and see the predicted price update instantly — with the maths shown on every page.

2. Fundamentals — Correlation

2.1 What correlation measures

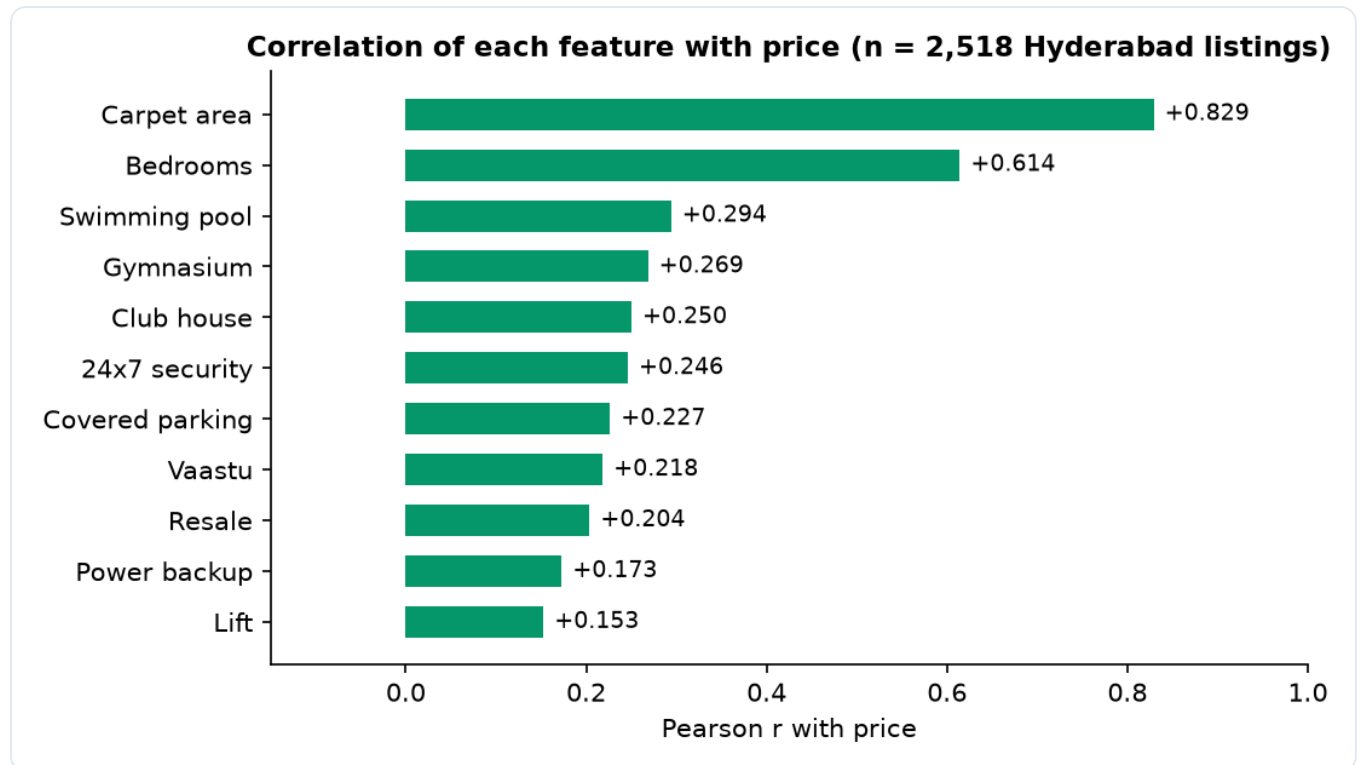
Correlation quantifies the strength and direction of a linear relationship between two variables. The **Pearson product-moment coefficient** r is defined as the covariance of the two variables divided by the product of their standard deviations:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{[\sum (x_i - \bar{x})^2 \cdot \sum (y_i - \bar{y})^2]}}$$

Here $\bar{x}$ and $\bar{y}$ are the sample means. The value of r always lies in $[-1, +1]$: $+1$ is a perfect positive linear relationship, -1 a perfect inverse one, and 0 means no *linear* relationship (a curved relationship may still exist). Common rules of thumb: $|r| \geq 0.7$ *strong*, ≥ 0.4 *moderate*, ≥ 0.2 *weak*, below that *very weak*.

Correlation is symmetric, unit-free, and sensitive to outliers and to the range of data sampled. Critically, it measures association in isolation — it cannot tell us how a variable behaves *once other variables are held constant*. That is the job of regression.

2.2 Correlation with price in our data



Feature	Pearson r with price	Strength	Interpretation
Carpet area (sq ft)	+0.829	Strong	Bigger homes cost more — the dominant single factor
Bedrooms (BHK)	+0.614	Moderate	More rooms → higher price, but area matters more
Swimming pool	+0.294	Weak	Amenities matter, but they mostly travel with area and locality
Gymnasium	+0.269	Weak	
Club house	+0.250	Weak	
24x7 security	+0.246	Weak	
Covered parking	+0.227	Weak	
Vaastu-compliant	+0.218	Weak	
Resale	+0.204	Weak	
Power backup · Lift	+0.173 · +0.153	Very weak	



Full correlation matrix of the 11 numeric features plus price; green is positive, red negative.

2.3 Why correlation alone is not enough

Three limitations motivate regression:

- **Confounding:** a swimming pool is correlated with price partly because pools are installed in large, expensive, high-grade projects. Correlation cannot separate “the pool adds value” from “pools come with expensive homes”.
- **No categorical locations:** locality is a categorical variable (48 areas) — correlation needs numeric inputs. Regression handles it with dummy variables.
- **No prediction:** correlation gives a strength, not a number. We want: “this exact home is worth approximately ₹X”.

3. Fundamentals — Regression

3.1 Simple linear regression

Linear regression models the expected price as a linear function of a predictor:

$$\text{price} = \beta_0 + \beta_1 \cdot \text{area} + \varepsilon$$

β_0 is the intercept (expected price when area = 0), β_1 is the slope (expected price increase per additional sq ft), and ε is the error. The coefficients are chosen by **Ordinary Least Squares (OLS)** — minimising the sum of squared residuals:

$$\min \hat{\beta}_0, \hat{\beta}_1 \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

R² (coefficient of determination) is the share of variance in price explained by the model: $R^2 = 1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$, where SS_{res} is the residual sum of squares and SS_{tot} the total sum of squares. **RMSE** (root mean squared error) reports the typical error in the original units (rupees) and is computed on data the model has never seen (the holdout set).

Area alone explains **69.3%** of price variance ($R^2 = 0.693$, RMSE ₹43.99 L) — a strong start, but the remaining 30% is where rooms, amenities and above all **location** live.

3.2 The coefficient interpretation shift

Correlation said *more bedrooms* → *higher price* ($r = +0.61$). But once area is held constant, the regression coefficient of bedrooms is **negative** (–₹16.0 L per extra bedroom in the 11-feature model): a 4-BHK with the same carpet area as a 3-BHK is priced *lower per square foot*. This is the classic lesson of multiple regression — coefficients measure an effect *controlling for the other variables*, and can legitimately differ from (even reverse) the raw correlation.

3.3 Multiple regression with dummy variables

We extend the model to all numeric features and encode the locality as **dummy (indicator) variables**. For 48 named localities (plus a residual “Other” group) we create 48 dummies, each 1 if the home sits in that area and 0 otherwise (one reference locality is dropped to avoid perfect multicollinearity):

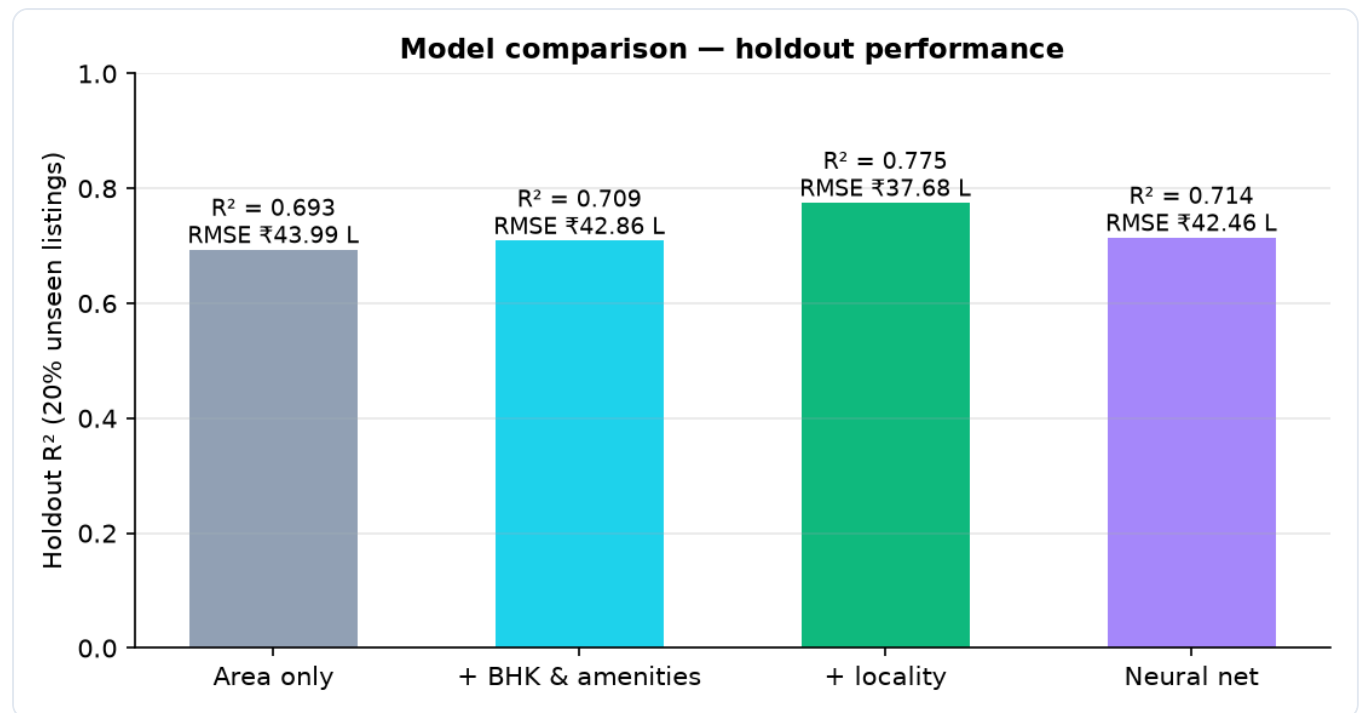
$$\text{price} = \beta_0 + \beta_1 \cdot \text{area} + \beta_2 \cdot \text{bedrooms} + \beta_3 \cdot \text{resale} + \beta_4 \dots \beta_{11} \cdot \text{amenities} + \sum \gamma_l \cdot 1(\text{locality} = l) + \varepsilon$$

Each γ_l is the locality’s price *premium* (or discount) relative to the reference area, after controlling for every feature. The fitted values are honest and interpretable:

Locality	Premium vs reference (₹)	Locality	Discount (₹)
Banjara Hills	+89.9 L	Tellapur	–21.8 L
Jubilee Hills	+84.1 L	Bandlaguda Jagir	–20.1 L
Madhapur	+41.7 L	Kollur Road	–18.7 L
Hitech City	+41.5 L	Patancheru	–21.0 L

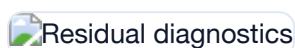
Key finding: location moves a home by up to ~₹1 Cr in either direction — more than any single feature. This is the project's central result.

3.4 Model progression and diagnostics



Model	Variables	Holdout R^2	Holdout RMSE
Area only	1	0.693	₹43.99 L
Multiple linear	11	0.709	₹42.86 L
Multiple + locality	11 + 48 dummies	0.775	₹37.68 L
Neural net (MLP)	11	0.714	₹42.46 L

Residual diagnostics (residuals vs fitted, Q–Q plot) confirm the expected shape for housing data: a fan-shaped variance that grows with price (heteroscedasticity) and a slightly fat right tail from premium listings — mild deviations from homoscedasticity, acceptable at $n = 2,518$ and reported honestly rather than “cleaned away”.



Left: residuals vs fitted (variance grows with price). Right: Q–Q plot (fat right tail).

The neural network (a multilayer perceptron, 32→16 hidden neurons, ReLU) beats plain linear regression on the same 11 features (R^2 0.714 vs 0.709, RMSE lower by ~1%), but cannot match the interpretable linear model *with locality* (0.775). That comparison — classical statistics vs machine learning, and the transparency of regression — is itself a finding of the project.

4. The Project

4.1 What was built

Everything above is packaged as a production-quality web application built with Next.js, TypeScript and Tailwind CSS, deployed on Vercel:

- **Live simulator** — sliders for carpet area and BHK, toggles for new/resale and 9 amenities, a 49-area locality dropdown; the fitted regression returns a price in ₹ with a ± 1 RMSE confidence band and a per-feature ₹ breakdown (“where the price comes from”).
- **All-in cost card** — adds Telangana stamp duty (4%) and registration (1%) to show the true purchase cost.
- **AI assistant** — natural language in, explained price out. A parser extracts features from text like “3 BHK, 1800 sq ft in Gachibowli with pool and gym”; the model computes the price; an LLM (OpenCode Go, glm-5.3-flash) writes the buyer-friendly explanation. An offline engine answers identically when no API key is present, so demos never break.
- **Interactive map** — all 2,518 listings plotted on an OpenStreetMap base, coloured by price, filterable by budget.
- **Maths pages** — correlation heatmaps, coefficient tables, residual diagnostics and a print-ready report page for submission.
- **Professional UI** — dark/light theme, glass surfaces, an SVG icon system and staggered animations.

PriceSim

Simulator

AI Assistant

Map

Insights

☀️

Get an estimate

☰

THE SIMULATOR

Hyderabad Price Simulator

Drag the sliders and pick a locality — the price updates instantly from a multiple-regression model fitted to 2,518 real Hyderabad listings (11 features + 48 locality dummies, $R^2 = 0.775$). The neural net gives a second opinion, and the closest real sales benchmark the estimate against reality.

Property details

Move anything — the estimate recalculates instantly.

Carpet area (sq ft)

1,500

Bedrooms (BHK)

3 BHK

Resale property

Swimming pool

Gymnasium ✓

Club house ✓

24x7 security ✓

Power backup ✓

Covered parking ✓

Lift ✓

Vaastu-compliant

Locality (46 areas of Hyderabad)

Gachibowli — median ₹1.26 Cr

Median is the middle sale price of 83 listings in this locality.

REGRESSION + LOCALITY

₹1.10 Cr

₹7,311/sqft · $R^2 = 0.775$

±₹37.7 L (1 RMSE)

₹72.0 L – ₹1.47 Cr

NEURAL NET (AI)

₹61.4 L

$R^2 = 0.714$ on holdout

±₹42.5 L (1 RMSE)

No locality input — pure maths

All-in cost with taxes

TELANGANA · APPROX

ESTIMATED PRICE

₹1.10 Cr

STAMP DUTY (4%)

₹4.4 L

REGISTRATION (1%)

₹1.1 L

TOTAL WITH TAXES

₹1.15 Cr

5% loading on the estimated price (approx. Telangana rates — adjust these numbers in lib/config.ts). Exact duty can differ by property value and buyer category.

Where the price comes from

Exact linear decomposition — each feature's contribution = coefficient × (value – average). In Gachibowli.

Locality

+₹3

Carpet area (sq ft)

₹-1

Club house

+₹1

Bedrooms (BHK)

₹-1

Covered parking

₹-0

Resale property

₹-0

24x7 security

+₹0

Gymnasium

₹-0

Lift

₹-0

Vaastu-compliant

+₹0

Power backup

+₹0

Swimming pool

+₹0

Sum of all contributions + baseline average home = predicted price. Rose bars drag the price down, green bars push it up.

10 most similar real listings

PRICE	AREA	BHK	RESALE	LOCALITY	SIMILARITY
₹62.2 L	1,555 sq ft	3	New	Alwal	0.07
₹48.6 L	1,214 sq ft	2	New	Miyapur	1.49
₹61.7 L	1,090 sq ft	2	New	Kukatpally	1.54
₹78.0 L	1,507 sq ft	3	New	Kokapet	2.01
₹93.0 L	1,470 sq ft	3	New	Chandanagar	2.01
₹79.6 L	1,435 sq ft	3	New	Kukatpally	2.02
₹81.0 L	1,435 sq ft	3	New	Kukatpally	2.02
₹82.4 L	1,585 sq ft	3	New	Nagole	2.02
₹1.15 Cr	1,592 sq ft	3	New	Kukatpally	2.02
₹81.0 L	1,620 sq ft	3	New	Other localities	2.02

k-NN over the same standardized features the model uses. Lower similarity score = closer match. These are actual asking prices from the dataset.

Real Estate Price Simulator

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Simulator

AI Assistant

Map

Explore

Correlation

Regression

Report

Team

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The simulator: live ₹ prediction for a 3-BHK in Banjara Hills with amenities, plus the tax-inclusive total.

4.2 The data

Property	Value
Source	2,518 real Hyderabad residential listings (public listing-site data)
Target	Asking price, ₹20 L – ₹16.5 Cr (median ₹77.5 L, mean ₹98.2 L)
Features	Carpet area, bedrooms (BHK), new/resale, 9 amenity flags, locality
Localities	48 named areas + “Other” (Banjara Hills median ₹2.99 Cr ... Nizampet ₹47.3 L)
Honesty note	Asking prices, not closed transactions — flagged inside the app whenever medians are shown

4.3 Team

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5. Conclusion

This project demonstrates the complete arc of a statistical study: **measure** (correlation), **model** (multiple regression), **validate** (holdout), **interpret** (coefficients and locality premia), and **communicate** (a live, interactive web application). The mathematics is classical and examinable — every coefficient, R^2 and RMSE in the app comes from the fitted models — while the delivery shows the topic in a form people can actually use: a prospective buyer can price a home in any part of Hyderabad in seconds, in rupees, and see exactly *why* the model says what it says.

References

- Dataset: public Hyderabad residential listings (99acres/MagicBricks scrape mirrors, GitHub public repositories).
- Analysis: Python 3.12 · pandas · NumPy · statsmodels · scikit-learn · matplotlib.
- Application: Next.js 16 · TypeScript · Tailwind CSS · Recharts · React-Leaflet · OpenCode Go (LLM).
- Deployment: Vercel (serverless) · GitHub Actions CI.